

Phase Transition and Instance-Dependent Complexity in Subset Sum Problem: An Empirical Study of Suffix-Sum Pruning

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Abstract

This study investigates the performance of backtracking algorithms for the Subset Sum Problem (SSP). By analyzing 2,400 experimental trials, we demonstrate that computational complexity is driven by data entropy rather than input size alone. Our results reveal a clear phase transition near critical density and show that structured inputs dramatically reduce computational cost due to effective pruning.

1 Introduction

The Subset Sum Problem (SSP) is a classical NP-complete problem. While worst-case complexity is exponential, practical performance often depends on input structure. This paper explores how data distribution (random vs structured) influences algorithmic behavior.

2 Methodology

We implemented a custom C++ engine on a modern ARM-based architecture. Experiments were conducted with $N = 34$ and varying bit-lengths $k \in 20, 30, 40, 50$, corresponding to density values $d = N/k$ in the range $[0.68, 1.70]$.

For each configuration, 300 trials were executed, totaling 2400 runs. We measured runtime and number of visited nodes (in logarithmic scale).

3 Results

The results show a clear phase transition in random datasets near $d \approx 1.0$. Runtime and node exploration increase sharply, reaching peak complexity. In contrast, structured datasets allow near-instant pruning, resulting in negligible computation time.

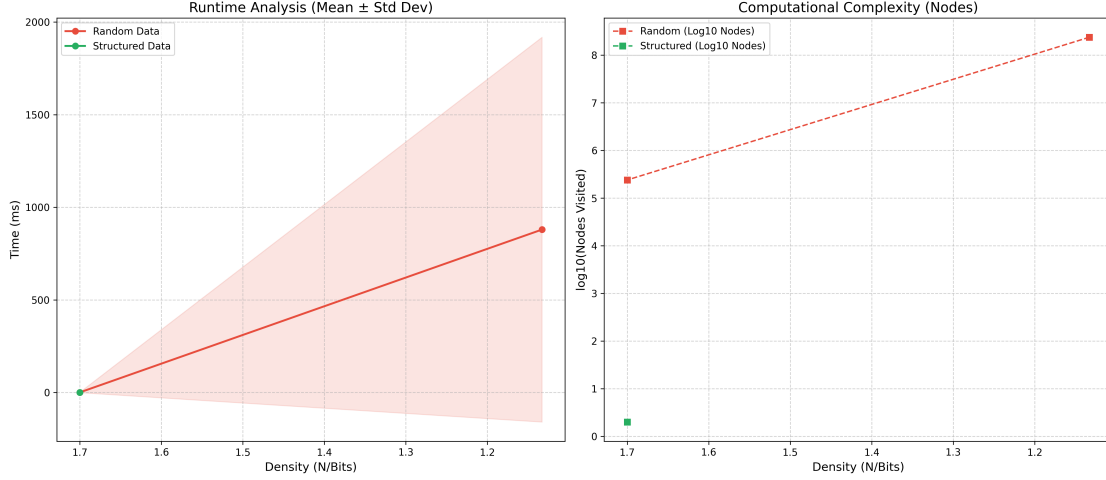


Figure 1: Runtime (left) and computational complexity (right) as a function of density. Random datasets exhibit a peak near critical density, while structured datasets remain computationally trivial due to aggressive pruning.

4 Discussion

The experiment confirms that algorithmic complexity in SSP is highly instance-dependent. While theoretical complexity remains exponential, practical performance varies significantly depending on data entropy and structure.

5 Conclusion

This work demonstrates that input structure plays a critical role in solving NP-complete problems in practice. Efficient pruning strategies can transform seemingly intractable problems into manageable ones when data exhibits patterns.